CHAPTER

2

Nomenclature and Terminology

For understanding the concepts of operative dentistry, it is essential that the clinician be thoroughly familiar with the technical terms used during the treatment. These terms also help in recording the patient's signs and symptoms, which can be useful during subsequent visits of the patient. Let us first be familiar with the notation systems, i.e. the systems used for denoting teeth in the arch.

The majority of the tooth notation systems fall into two categories:

A. Systems having similar notation in each segment
B. Systems having different notation in each segment

Tooth notation systems were originally developed for designating permanent teeth, later supplementary systems for the temporary dentition were established; often by adding to the symbols used in the basic systems.

In the description of various systems that follows, the method used for denoting temporary teeth is given second to the method used for denoting permanent teeth.

A. SYSTEMS HAVING SIMILAR NOTATION IN EACH SEGMENT

System 1: The oldest method used in this group is 'Zsigmondy system' (Zsigmondy, 1861). The central incisor of each segment is given the number '1' and the numbers then run in the distal direction. The segments are shown as the patients' upper right, upper left, lower right and lower left segments, respectively.

The Zsigmondy system is frequently employed both in Europe and America. It is also common in Australia and Japan. In English speaking countries, it is generally known as Palmer's notation (Palmer, 1891).

The temporary teeth are indicated merely by altering the Arabic numerals to Roman ones.

Zsigmondy's system (permanent teeth)

R	8	7	6	5	4	3	2	1	1	2	3	4	5	6	7	8	L
	8	7	6	5	4	3	2	1	1	2	3	4	5	6	7	8	

Sometimes temporary teeth are designated with the letters a–e or with capital letter A–E instead of Roman numerals I–V or letter D (deciduous) placed after the number of the tooth or a small 'd' is placed before the number of the tooth. The temporary teeth may also be indicated by the addition of the letter 'm' after the number of the tooth.

Different variants for temporary teeth

R	V	IV	III	II	I	I	II	III	IV	V	L
R	e	d	c	b	a	a	b	c	d	e	L
	e	d	c	b	a	a	b	c	d	e	
R	E	D	C	B	A	A	B	C	D	E	L
	E	D	C	B	A	A	B	C	D	E	
R	5D	4D	3D	2D	1D	1D	2D	3D	4D	5D	L
	5D	4D	3D	2D	1D	1D	2D	3D	4D	5D	
R	d5	d4	d3	d2	d1	d1	d2	d3	d4	d5	L
	d5	d4	d3	d2	d1	d1	d2	d3	d4	d5	
R	5m	4m	3m	2m	1m	1m	2m	3m	4m	5m	L
	5m	4m	3m	2m	1m	1m	2m	3m	4m	5m	

System 2: This system employs numerals 1 to 8 for enumerating the permanent teeth and A to E for enumerating the temporary teeth. It is exactly opposite to the above mentioned Zsigmondy's system. In this system, notation begins with number 1 for the third molar and with the letter A for the second temporary molar. It ends with 8 and E for the permanent and deciduous central incisors, respectively.

R	1	2	3	4	5	6	7	8	8	7	6	5	4	3	2	1	L
	1	2	3	4	5	6	7	8	8	7	6	5	4	3	2	1	
R	A	B	C	D	E	E	D	C	B	A	L						
	A	B	C	D	E	E	D	C	B	A							

System 3: These systems do not make use of numerals, but designate the teeth (starting from the central

incisor) as: I_1, I_2, C, P_1, P_2, M_1, M_2 and M_3, i.e. the initial letters of their respective Latin names. Temporary teeth are shown with small letters, sometimes supplemented with the letter d (deciduous) preceding the letter symbol.

Permanent teeth

R	M_3	M_2	M_1	P_2	P_1	C	I_2	I_1	I_1	I_2	C	P_1	P_2	M_1	M_2	M_3 L
	M_3	M_2	M_1	P_2	P_1	C	I_2	I_1	I_1	I_2	C	P_1	P_2	M_1	M_2	M_3

Temporary teeth

R	dm_2	dm_1	dc	di_2	di_1	di_1	di_2	dc	dm_1	dm_2 L
	dm_2	dm_1	dc	di_2	di_1	di_1	di_2	dc	dm_1	dm_2

The system 3 is employed mainly in Holland.

System 4: This system was invented by Dane, Haderup (1887, 1891). According to this system, the teeth are numbered in each segment starting with number 1 for the central incisor. The teeth are then numbered from 1 to 8 in a distal direction. The tooth numbers in the upper jaw are combined with a plus sign, those in the lower jaw with a minus sign. These signs are placed to the right of the numeral, if the tooth is situated on the right side of the jaw and to the left of the numeral, if the tooth is situated on the left side of the jaw. Temporary teeth were originally shown by the addition of the letter L placed before the numeral. After a few years, this was altered to another form, whereby 0, also placed before the numeral, substituted the letter L. In central Europe, the temporary teeth are indicated by Roman numerals (without the addition of 0) in conjunction with + and – signs.

Haderup's system (permanent teeth)

R	8+	7+	6+	5+	4+	3+	2+	1+	+1	+2	+3	+4	+5	+6	+7	+8 L
	8–	7–	6–	5–	4–	3–	2–	1–	–1	–2	–3	–4	–5	–6	–7	–8

Variants one for temporary teeth

R	05+	04+	03+	02+	01+	+01	+02	+03	+04	+05 L
	05–	04–	03–	02–	01–	–01	–02	–03	–04	–05

Variants two for temporary teeth

R	V+	IV+	III+	II+	I+	+I	+II	+III	+IV	+V L
	V–	IV–	III–	II–	I–	–I	–II	–III	–IV	–V

Haderup's system is mainly used in Sweden, Denmark, Norway, Finland and Iceland.

System 5: In this system, the incisors (I), canines (C), premolars (P) and molars (M) are indicated in the following way:

I_1, I_2, C, P_1, P_2, M_1, M_2 and M_3.

The upper jaw is indicated by the letter s (superior) and the lower jaw by the letter i (inferior) placed immediately after the index numeral and followed by d or s (dexter and sinister) for the right and left side, respectively. The method is the same for the temporary teeth, the difference being that small letters are used to represent the teeth, i.e. i_1, i_2, c, m_1, m_2. The system is mainly used in Holland.

System 6: This system is similar to System 5, having the same letter and index numeration for the teeth; however, the segment is identified by the position of the index in relation to the alphabetical letter of the tooth. For the upper right segment, the index is placed higher and to the left of the alphabetical symbol of the tooth; for the upper left segment, higher and to the right of it. For the lower right segment, the index is placed lower and to the left of the alphabetical symbol of the tooth; for the left segment, lower and to the right of it. This system is mainly used in South Africa.

R	^{3}M	^{2}M	^{1}M	^{2}P	^{1}P	^{1}C	^{2}I	^{1}I	I^1	I^2	C^1	P^1	P^2	M^1	M^2	M^3 L
	$_3$M	$_2$M	$_1$M	$_2$P	$_1$P	$_1$C	$_2$I	$_1$I	I_1	I_2	C_1	P_1	P_2	M_1	M_2	M_3

System 7: In this system, the teeth are numbered 1–8 from the central incisor to the last molar. The teeth of the upper jaw are indicated by a capital D (droite) or G (gaucha) for the right or left side, respectively and is placed immediately prior to the number of the tooth. The teeth of the lower jaw are indicated by a small d or g.

Temporary teeth are indicated by substituting Roman numerals for the Arabic ones. This system has limited use, mainly in France and Romania.

R	D8	D7	D6	D5	D4	D3	D2	D1	G1	G2	G3	G4	G5	G6	G7	G8 L
	d8	d7	d6	d5	d4	d3	d2	d1	g1	g2	g3	g4	g5	g6	g7	g8

B. SYSTEMS HAVING DIFFERENT NOTATION IN EACH SEGMENT

Several systems which employ different notations for the teeth in different segments are in use. The most common of these systems are 'Army system', 'Navy system', 'Universal system' and the 'Bosworth system'. These are as follows:

System 8:

Army System (permanent teeth)

R	8	7	6	5	4	3	2	1	1	2	3	4	5	6	7	8 L
	16	15	14	13	12	11	10	9	9	10	11	12	13	14	15	16

The temporary teeth in Army system are indicated by placing the letter 'T' (Temporary) in front of the number of the corresponding temporary tooth.

Variant of Army system for temporary teeth

5T	4T	3T	2T	1T	1T	2T	3T	4T	5T
13T	12T	11T	10T	9T	9T	10T	11T	12T	13T

System 9:

Navy System (permanent teeth)

R																L
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32

The first variant of the Navy system for temporary teeth implies as: The second temporary molar on the right side of the upper jaw retains the number '4' and the second temporary molar on the left-hand side of the lower jaw retains the number '29'. The first variant shows deciduous teeth by putting a ring round the number of tooth.

Variant (one) of the Navy system for temporary teeth

R										L
	④	⑤	⑥	⑦	⑧	⑨	⑩	⑪	⑫	⑬
	⑳	㉑	㉒	㉓	㉔	㉕	㉖	㉗	㉘	㉙

In the second variant, this is shown by the addition of '1/2' to the number of permanent tooth (4½, 29½).

Second variant of the Navy system for temporary teeth

R										L
	4½	5½	6½	7½	8½	9½	10½	11½	12½	13½
	20½	21½	22½	23½	24½	25½	26½	27½	28½	29½

The third variant of Navy system implies as: The temporary teeth of the upper jaw are numbered from I–X, beginning with the second temporary molar on the right side, and those in the lower jaw from XI–XX.

Third variant of the Navy system for temporary teeth

R											L
	I	II	III	IV	V	VI	VII	VIII	IX	X	
	XI	XII	XIII	XIV	XV	XVI	XVII	XVIII	XIX	XX	

The fourth variant is to use letters A–J for the temporary teeth of the upper jaw and letters K–T for the lower jaw.

Fourth variant of the Navy system for temporary teeth

R											L
	A	B	C	D	E	F	G	H	I	J	
	K	L	M	N	O	P	Q	R	S	T	

System 10:

Bosworth system (permanent teeth)

R																L
	8	7	6	5	4	3	2	1	1	2	3	4	5	6	7	8
	H	G	F	E	D	C	B	A	A	B	C	D	E	F	G	H

With the Bosworth system, the temporary teeth are indicated in the following way:

Variant of Bosworth system for temporary teeth

R											L
	D5	D4	D3	D2	D1	D1	D2	D3	D4	D5	
	DE	DD	DC	DB	DA	DA	DB	DC	DD	DE	

System 11:

Universal System (permanent teeth)

R																L
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17

Figure 2.1 clarifies universal system for permanent teeth.

Variant of Universal system for primary teeth

R											L
	A	B	C	D	E	F	G	H	I	J	
	T	S	R	Q	P	O	N	M	L	K	

Figure 2.2 clarifies universal system for primary teeth.

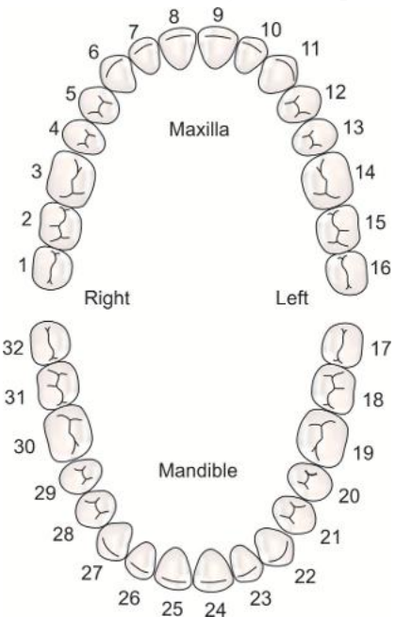


Fig. 2.1: Universal system for permanent teeth

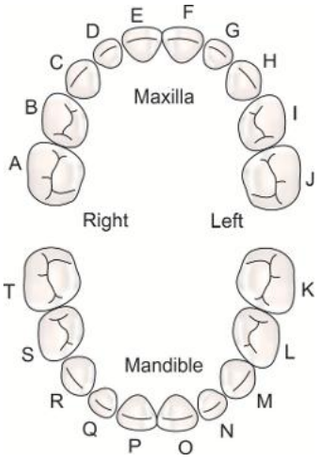


Fig. 2.2: Universal system for primary teeth

For each system, two principles are followed for designating temporary teeth; (i) retain the designation of the first five teeth in each segment and indicate the temporary teeth by making some addition to the symbol of the tooth; (ii) retain the method for designating the teeth, but substitute for the numbers of the permanent teeth. Substitutions can be in the form of alphabetical letters or different numerals. The temporary teeth may have different symbols from the permanent teeth.

FDI SYSTEM

The Federation Dentaire Internationale (FDI) has evolved two digit system, which is accepted all around the world. This tooth notation system was developed by Dr. J. Viohl, which makes visual sense, cognitive sense and computer sense. It fulfils the following basic requirements (Figs 2.3 to 2.6).

- Simple to teach and understand
- Readily communicable in print and telephone
- Easy to speak in conversation and dictation
- Easy to enter into a computer
- Easily adaptable to standard charts used in general practice

The FDI two digit system identifies each of the 32 permanent teeth with a two digit number, the first digit indicating the quadrant (1 to 4, starting from upper right quadrant clockwise to lower right quadrant) and the second digit indicating the tooth type (1 to 8, starting from central incisor to the third molar). The 20 primary teeth are represented in a similar fashion: Quadrant as 5 to 8 and tooth type as 1 to 5.

FDI system (permanent teeth)																
Maxillary																
R	18	17	16	15	14	13	12	11	21	22	23	24	25	26	27	28 L
	48	47	46	45	44	43	42	41	31	32	33	34	35	36	37	38
Mandibular																

FDI system (primary teeth)										
Maxillary										
R	55	54	53	52	51	61	62	63	64	65 L
	85	84	83	83	81	71	72	73	74	75
Mandibular										

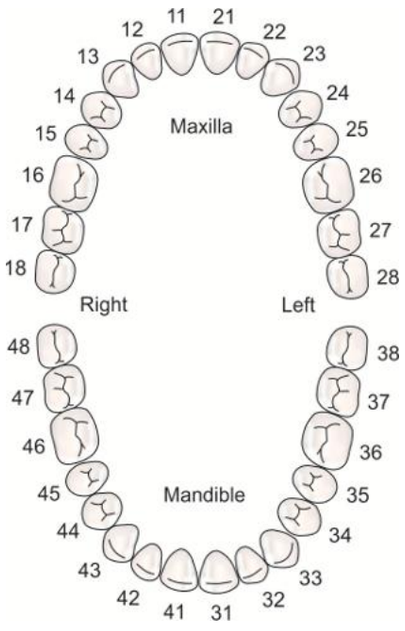


Fig. 2.3: FDI system for permanent teeth

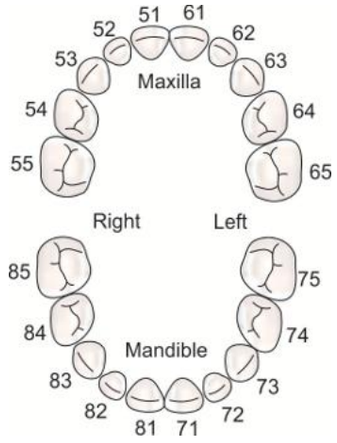


Fig. 2.4: FDI system for primary teeth

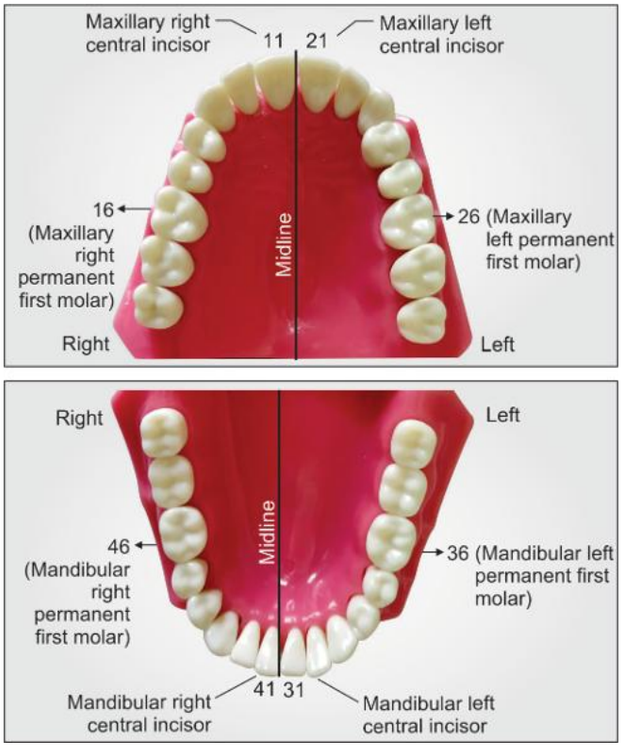


Fig. 2.5: FDI system (maxillary-mandibular teeth)



Fig. 2.6: FDI system (permanent teeth)

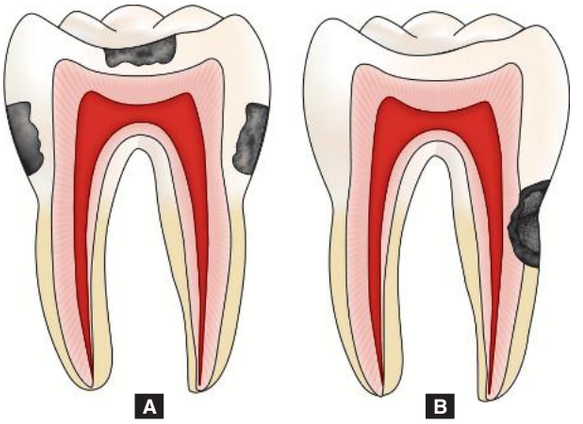


Fig. 2.7: Caries on different surfaces of teeth (diagrammatic): **(A)** In pits and fissures and on smooth surfaces; **(B)** On root surfaces

TERMINOLOGY RELATED TO CARIES

Dental caries is an infectious and microbiological disease that results in localized dissolution and destruction of the calcified tissues of the tooth.

Caries involving various tooth surfaces is denoted by that particular surface name used as a prefix before the word 'caries', e.g. caries on the mesial surface of the tooth is referred to as mesial caries. Caries involving two surfaces say mesial and occlusal is referred to as mesio-occlusal (MO) caries. Caries involving three surfaces say mesial, occlusal and distal is referred to as mesio-occluso-distal (MOD) caries.

Types of Caries

A. According to the location
- a. *Primary caries*
  - i. Pit and fissure caries
  - ii. Smooth surface caries
  - iii. Root caries
  - iv Residual caries
- b. *Secondary/recurrent caries*

B. According to the amount of tooth involvement
- a. Incipient caries
- b. Advanced/cavitated caries

C. According to the rate of progression
- a. Acute caries
- b. Chronic caries

A. According to the Location

a. Primary Caries

The original carious lesion in the tooth is referred to as the primary caries. Depending upon its location on the tooth, the pattern of caries progression varies and also influences the cavity preparation. Caries can be located in the pits and fissures, on smooth surfaces and root surfaces (Fig. 2.7A, B).

i. ***Pit and fissure caries:*** Caries beginning in the pits and fissures of teeth is referred to as pit and fissure caries (Fig. 2.8). Pits and fissures are those areas of the teeth where there is imperfect coalescence of developmental enamel lobes. Because of incomplete fusion of enamel, these areas are susceptible to food impaction and hence caries.



Fig. 2.8: Pit and fissure caries

ii. ***Smooth surface caries:*** Smooth surface caries begin on relatively smooth surfaces of the tooth (Fig. 2.9A, B). Unlike pit and fissure caries, smooth surface caries initially involves a larger area of enamel on its outer surface.

iii. ***Root caries:*** Root surface caries also referred to as senile caries, begins on the roots of teeth that have been exposed to the oral environment (Fig. 2.10A,B). The progression of root caries is rapid and the prevalence is also increasing for the last few years.

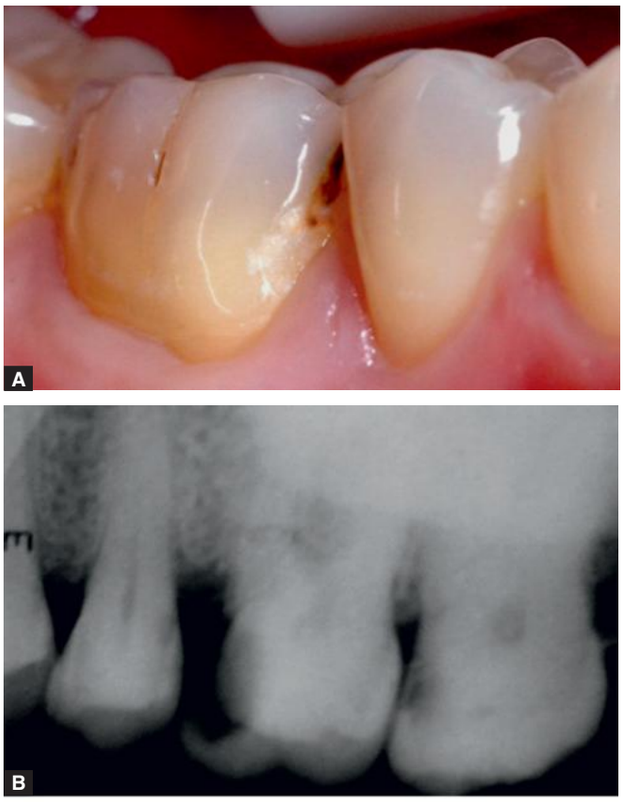


Fig. 2.9: Smooth surface caries: **(A)** Clinical; **(B)** Radiographic

iv. ***Residual caries:*** Caries that remains after the complete cavity preparation is referred to as residual caries, which may have been left behind either intentionally by the operator or by accident (Fig. 2.11).

b. *Secondary Caries*

Secondary caries, also referred to as recurrent caries, begins around or beneath the restoration (Fig. 2.12A,B). Its occurrence is suggestive of an improper seal between the tooth and restoration where micro-leakage may occur predisposing to the development of caries.

B. According to the Amount of Tooth Involvement

a. *Incipient Caries*

Incipient caries, also referred to as initial caries, is the beginning of caries activity. The lesion is evident as a white opaque area on the surface of the enamel (Fig. 2.13). Surfaces inflicted by incipient caries are fairly hard and only minor surface roughening may be present. Unlike hypoplastic white areas which are visible irrespective of whether the tooth surface is dry or wet, incipient caries becomes visible only when the tooth surface is dried. At this stage, the lesion is in a reversible state (can be remineralized).

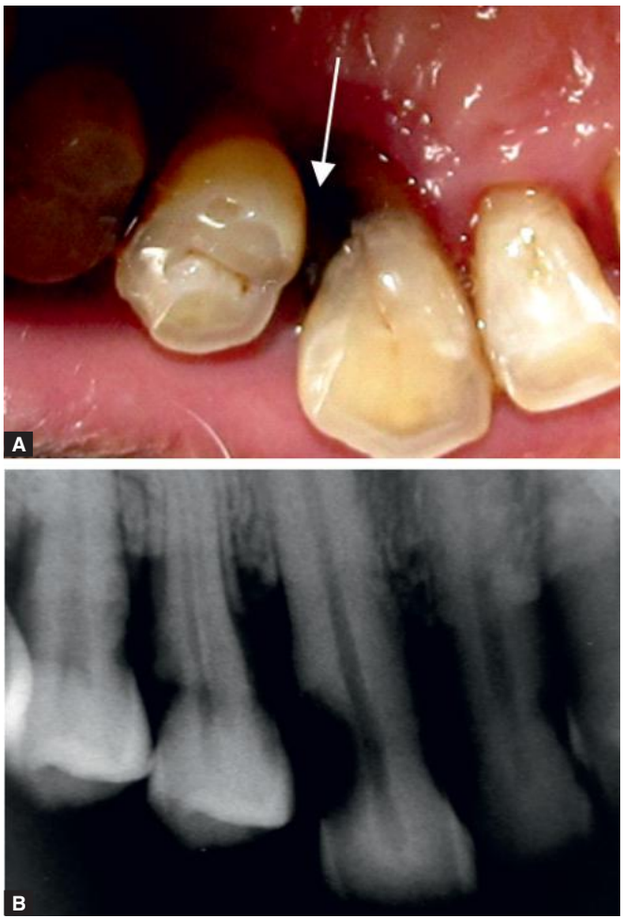


Fig. 2.10: Root caries: **(A)** Clinical (arrow); **(B)** Radiographic



Fig. 2.11: Residual caries



Fig. 2.12: Secondary caries: **(A)** Clinical; **(B)** CBCT image



Fig. 2.13: Incipient caries (arrow)

b. Advanced/Cavitated Caries

Caries that has progressed to the dentinoenamel junction is called advanced caries (Fig. 2.14). When the overlying enamel breaks down, the lesion is known as cavitated caries. At this stage, the lesion cannot be remineralized and requires cavity preparation and restoration for treatment.

C. According to the Rate of Progression

a. Acute Caries

Caries of a rapid onset and spread is referred to as acute caries, e.g. nursing bottle caries or rampant caries (Fig. 2.15). Lesions are light yellow in color, soft and highly infectious.

b. Chronic Caries

Caries of a slower onset and spread is referred to as chronic caries. The lesion is hard and dark brown to black in color (Fig. 2.16). The dark color is because of external staining.



Fig. 2.14: Advanced caries



Fig. 2.15: Acute caries

DEFECTS IN THE TOOTH STRUCTURE

a. ***Attrition:*** The loss of tooth structure occurring as a result of frictional contact between the opposing teeth (Fig. 2.17).
b. ***Abrasion:*** Lesions formed as a result of wearing away of the tooth substance because of grinding, rubbing or scraping caused by external mechanical means; the repeated contact of teeth with foreign objects or substances may lead to abrasion (Fig. 2.18).
c. ***Abfraction:*** These are wedge-shaped defects in the cervical region of the tooth, believed to be the result of tensile stress concentrated in this area consequent to occlusal forces in some remote areas (Fig. 2.19).
d. ***Erosion:*** Lesions formed as a result of dissolution of tooth structure subsequent to chemical attack of



Fig. 2.16: Chronic caries



Fig. 2.17: Attrition



Fig. 2.18: Abrasion

either endogenous or exogenous origin, or combined chemico-mechanical attack. Depending upon the source of chemicals (usually acids), erosion may be intrinsic or extrinsic (Fig. 2.20 A, B).



Fig. 2.19: Abfraction

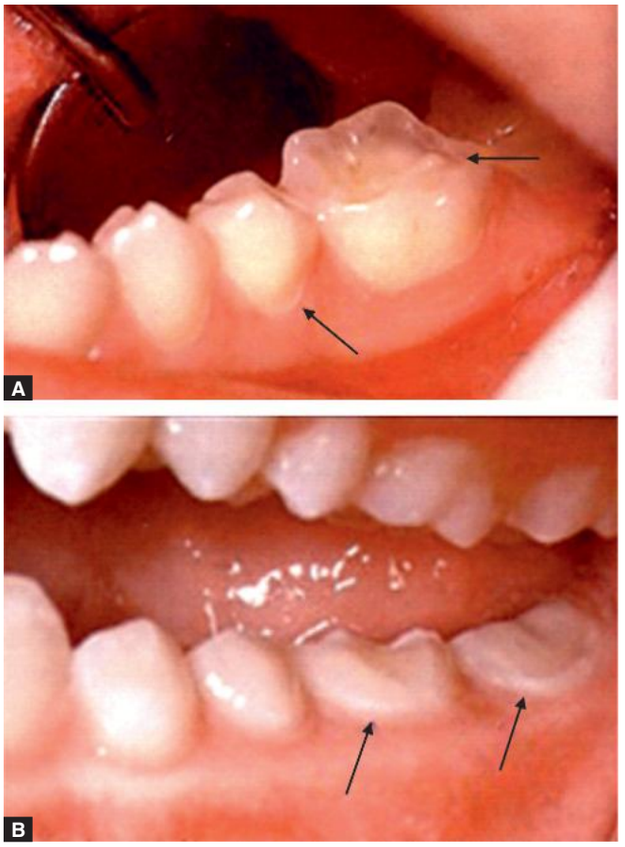


Fig. 2.20: Erosion (arrow)

e. ***Fracture:*** Any break in the continuity of the tooth surface is referred to as a fracture (Fig. 2.21).

f. ***Enamel hypoplasia:*** Defective formation of the enamel that may occur because of injury to the ameloblasts is called enamel hypoplasia. Hypoplasia is seen as deformed pits and grooves on the tooth surface; may be as opaque white to light brown area with smooth hard surface (Fig. 2.22).

g. ***Hypocalcification:*** Defective mineralization of the tooth is referred to as hypocalcification; seen as an



Fig. 2.21: Fracture in maxillary central incisor



Fig. 2.22: Generalized enamel hypoplasia



Fig. 2.23: Enamel hypocalcification

opaque white area that is soft to an explorer touch (Fig. 2.23).

h. ***Amelogenesis imperfecta:*** This is a heredity condition in which enamel is defective. It may be either hypoplastic or hypocalcified or both. The teeth may appear normal or extremely unsightly (Fig. 2.24).

i. ***Dentinogenesis imperfecta:*** This is a heredity condition in which only dentin is defective but enamel is normal. Enamel is loosely attached to the underlying dentin and is lost early in the life (Fig. 2.25).



Fig. 2.24: Amelogenesis imperfecta



Fig. 2.25: Dentinogenesis imperfecta

NOMENCLATURE RELATED TO TOOTH SURFACES

Different surfaces of the tooth (Fig. 2.26A, B) are named according to their adjoining anatomic structures (Table 2.1), viz. the surfaces facing towards cheeks are buccal and the surface towards palate/tongue is palatal/lingual. The surfaces towards incisors are mesial surfaces and the surfaces towards molars are distal surfaces.

Table 2.1: Different surfaces of the tooth

Facial surfaces	Labial surface	Facing towards the lip
	Buccal surface	Facing towards the cheeks
Lingual surface		Facing towards the tongue
Palatal surface		Facing towards the palate
Mesial surface		Facing towards the midline
Distal surface		Facing away from the midline
Incisal surface		Functioning edges of the incisors and canines
Occlusal surface		Functioning/masticating surfaces of the premolars and molars
Cervical area		Area of the tooth related to the cervical line (necks of teeth)
Gingival area		Area of the tooth close to the gingiva

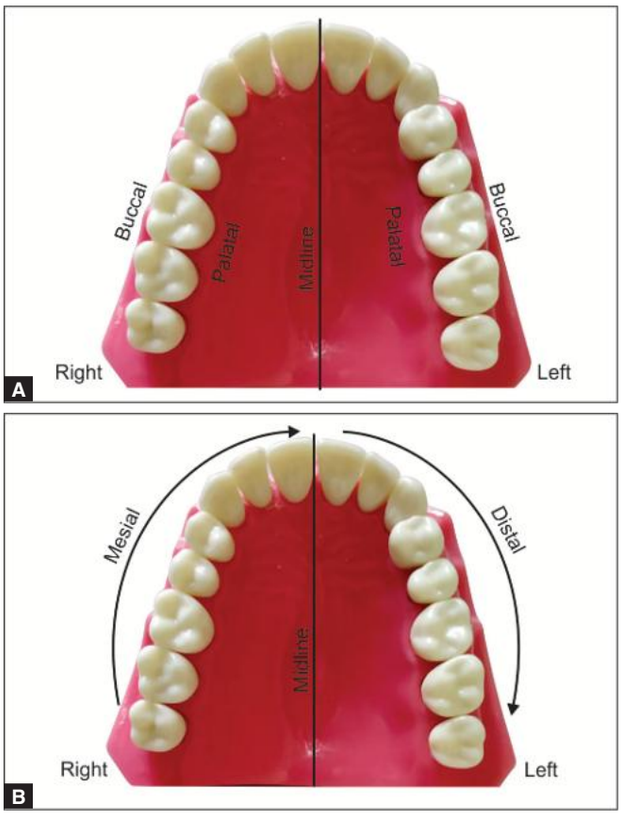


Fig. 2.26: Diagrammatic representation of surfaces of teeth: **(A)** Buccal (facing cheeks), palatal (facing palatal side); **(B)** Mesial (towards incisors); distal (towards molars)

ANATOMICAL/CLINICAL CROWN

Anatomical crown is the part of tooth, covered with enamel; whereas, clinical crown in the part of tooth, visible in the oral cavity (Fig. 2.27A, B).

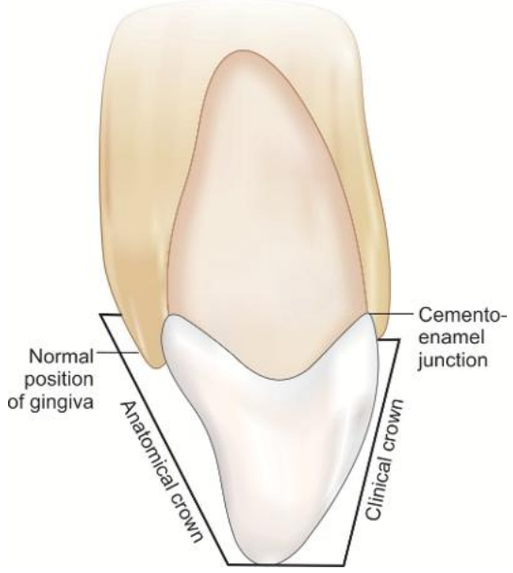


Fig. 2.27A: Anatomical crown and clinical crown (normal position of gingiva)

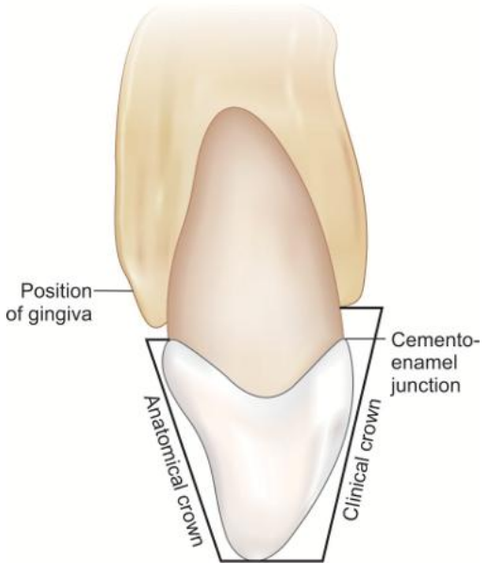


Fig. 2.27B: Anatomical crown and clinical crown (receded position of gingiva)

NOMENCLATURE RELATED TO CAVITY PREPARATION

a. ***Cavity:*** A cavity refers to a defect in enamel or both in enamel and dentin or in cementum, subsequent to the destruction caused by dental caries.

b. ***Cavity preparation:*** Cavity preparation is the alteration of defective/diseased tooth structure by dental surgical procedures and subsequently shaping the remaining enamel and dentin to receive an appropriate restorative material.

   i. *Intracoronal cavity preparation:* Cavity that is prepared in the interior of the tooth is referred to as the intracoronal cavity preparation.

   ii. *Extracoronal cavity preparation:* Cavity preparation that involves the external surfaces of the tooth is referred to as the extracoronal cavity preparation.

Simple Cavities

Cavities involving only one surface of the tooth are called simple cavities (Fig. 2.28), e.g. mesial (M), distal (D), facial (F), lingual (L), occlusal (O) and incisal (I) cavities.

Compound Cavities

Cavities involving two adjoining surfaces of the tooth are called compound cavities (Fig. 2.29). For example:

MO – Mesio-occlusal cavity
MB – Mesio-buccal cavity
ML – Mesio-lingual cavity
DO – Disto-occlusal cavity
DB – Disto-buccal cavity
DL – Disto-lingual cavity



Fig. 2.28: Simple cavity



Fig. 2.29: Compound cavity

Complex Cavities

Cavities involving more than two adjoining surfaces of the tooth are called *complex cavities* (Fig. 2.30), for example:

Mesio-occluso-distal cavity (MOD cavity)
Mesio-inciso-distal cavity (MID cavity)
Facio-occluso-lingual cavity (FOL cavity)



Fig. 2.30: Complex cavity

Pit and Fissure Cavities

Cavities involving pits and fissures of anterior and posterior teeth are referred to as pit and fissure cavities.

Smooth Surface Cavities

Cavities involving the smooth surfaces of the teeth are referred to as smooth surface cavities.

WALLS IN A CAVITY PREPARATION

a. ***Wall:*** Any surface of the cavity is referred to as a wall.
   i. *Internal wall:* Surface of a prepared cavity that does not extend to the exterior of the tooth is referred to as an internal wall.
   ii. *External wall:* Surface of a prepared cavity that extends onto the exterior of the tooth is referred to as an external wall.
   iii. *Enamel wall:* That portion of the cavity wall which is composed of enamel is called an enamel wall.
   iv. *Dentin wall:* That portion of the cavity wall which is composed of dentin is called a dentin wall.

b. ***Floor/seat of the cavity:*** Any cavity wall that is flat and perpendicular to the forces directed occlusogingivally is referred to as a floor/seat of the cavity, e.g. pulpal and gingival walls.

c. ***Step:*** An auxiliary extension of the main cavity on to an adjoining surface is referred to as a step. For example, in a Class III cavity with a lingual dovetail, the lingual dovetail is referred to as a lingual step.

d. ***Dentinoenamel junction (DEJ):*** The line of union between enamel and dentin is referred to as the dentinoenamel junction.

e. ***Cementoenamel junction (CEJ):*** The line of union between enamel and cementum is referred to as the cementoenamel junction; also known as the cervical line.

Different walls of a cavity are named according to the surfaces towards which they face (Table 2.2).

Walls in an Occlusal Cavity (Class I) (Fig. 2.31)

- Facial wall
- Lingual wall
- Mesial wall
- Distal wall
- Pulpal wall

Walls in a Proximo-occlusal Cavity (Class II) (Fig. 2.32)

Occlusal portion

- Facial wall
- Lingual wall

Table 2.2: Different walls in a cavity preparation

Facial walls	Labial wall	Wall facing towards the lips
	Buccal wall	Wall facing towards the cheeks
Lingual/palatal walls		Walls facing towards the tongue and palate, respectively
Incisal/occlusal walls		Walls facing towards the incisal and occlusal portions of the tooth, respectively
Mesial/distal walls		Walls facing towards the mesial and distal aspects of the tooth, respectively
Axial wall		Wall parallel to the long axis of the tooth and facing the pulp
Pulpal wall		Wall perpendicular to the long axis of the tooth in cavities present on the occlusal surface/incisal edges of the teeth, facing the pulp
Gingival wall		Wall facing the gingiva
Subpulpal wall		When the roof of the pulp chamber is removed, the floor of the pulp chamber left behind is referred to as the subpulpal wall

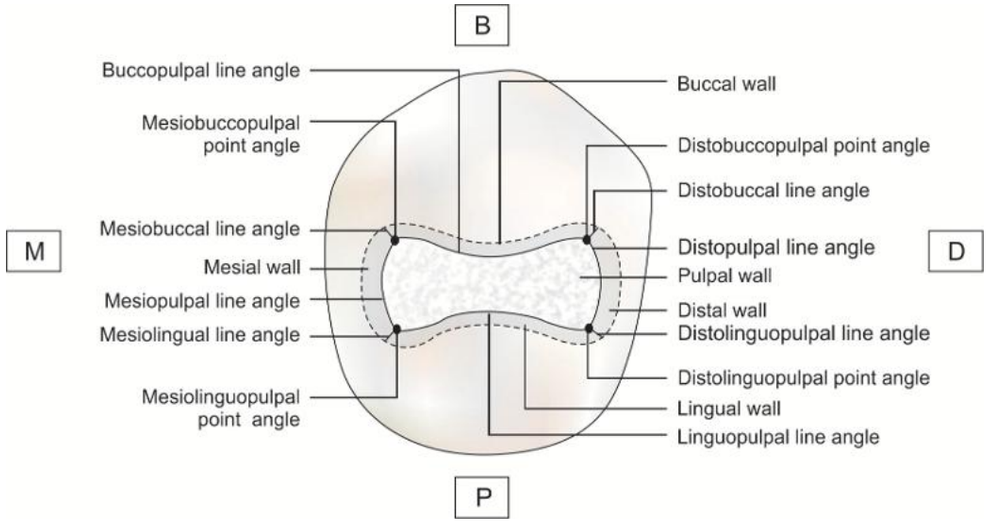


Fig. 2.31: Walls and angles in an occlusal Class I cavity (maxillary first premolar). M: mesial, D: Distal, B: Buccal, P: Palatal

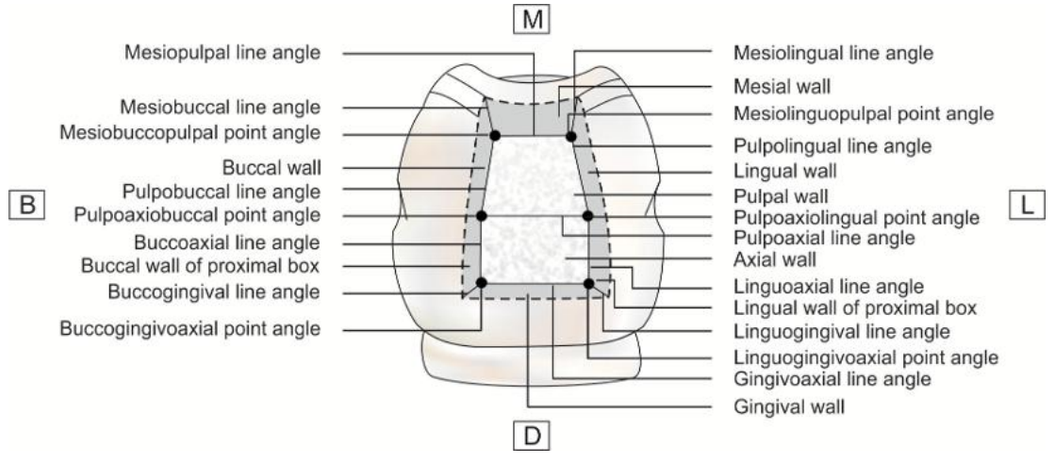


Fig. 2.32: Walls and angles in a Class II cavity (mandibular first molar). B: Buccal, L: Lingual, M: Mesial, D: Distal

- Mesial or distal wall (any one of these, depending on which surface the cavity is present)
- Pulpal wall

Proximal portion

- Facioproximal wall
- Linguoproximal wall
- Gingival wall
- Axial wall

Walls in a Proximal Cavity (Class III) (Fig. 2.33)

- Facial wall
- Lingual wall
- Gingival wall
- Axial wall

Walls in a Proximoincisal Cavity (Class IV)

Incisal step

- Facial wall
- Lingual wall
- Mesial or distal wall (any one of these depending on which surface the cavity is present)
- Pulpal wall

Proximal portion

- Facioproximal wall
- Linguoproximal wall
- Gingival wall
- Axial wall

Walls in Facial and Lingual Cavities (Class V) (Fig. 2.34)

- Mesial wall
- Distal wall

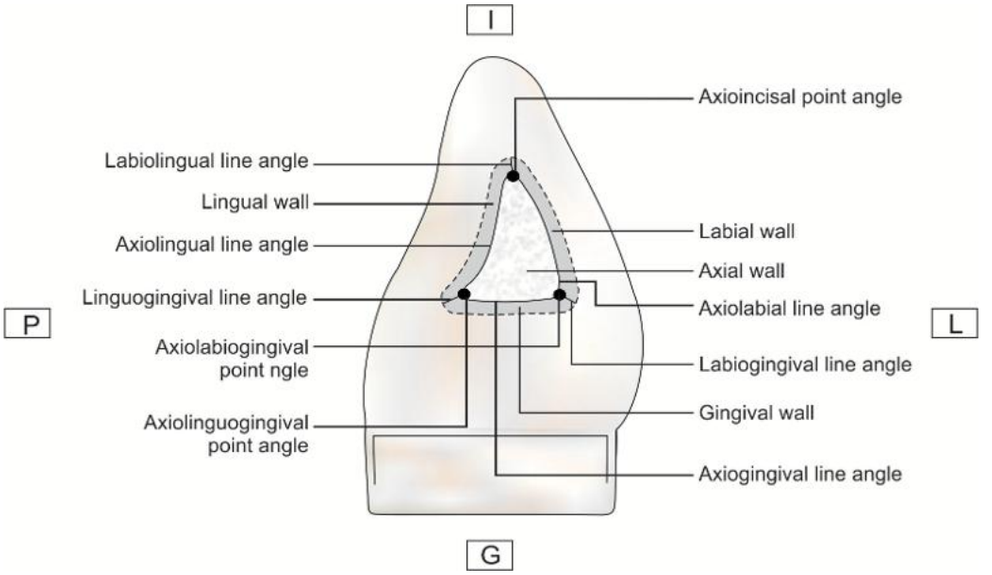


Fig. 2.33: Walls and angles in a Class III cavity (maxillary canine). P: Palatal, I: Incisal, G: Gingival, L: Lingual

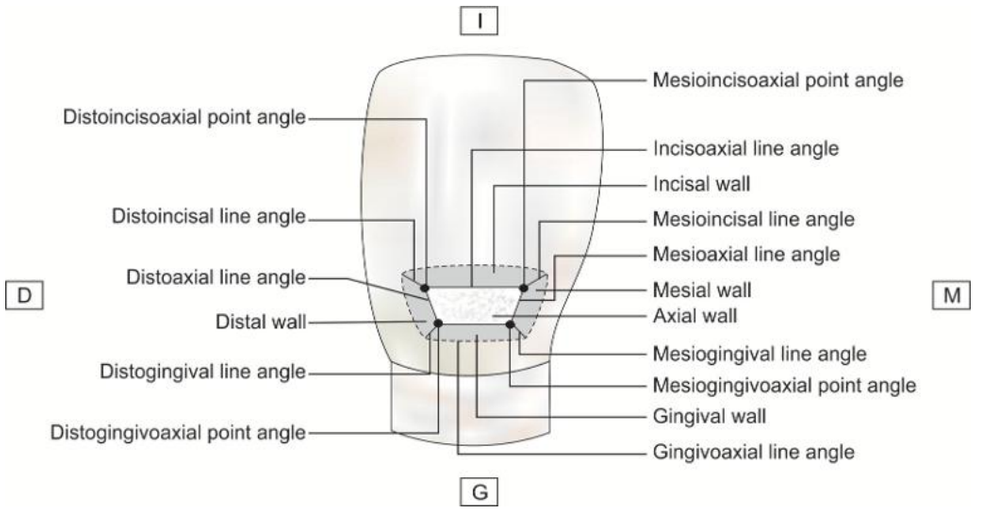


Fig. 2.34: Walls and angles in a Class V cavity (maxillary central incisor). D: Distal, I: Incisal, G: Gingival, M: Mesial

- Occlusal/incisal wall
- Gingival wall
- Axial wall

Walls in a Mesio-occluso-distal Cavity

Mesial proximal box
- Facioproximal wall
- Linguoproximal wall
- Gingival wall
- Axial wall

Distal proximal box
- Facioproximal wall
- Linguoproximal wall
- Gingival wall
- Axial wall

Occlusal portion
- Facial wall
- Lingual wall
- Pulpal wall

How to Combine Terms?

When one word is to be formulated by combining the names of two or more surfaces/walls, the 'al' ending of the prefix word is changed to an 'o', e.g. if mesial and occlusal is to be combined, the word so formulated is mesio-occlusal and similarly if mesial, distal and occlusal have to be combined, the word so formulated is mesio-disto-occlusal.

ANGLES IN A CAVITY PREPARATION

Junction of two or more surfaces of a prepared cavity is referred to as 'angle':

i. ***Line angle:*** The angle formed at the junction of two adjoining walls in a cavity preparation is referred to as a line angle. For example, angle formed between a mesial wall and a pulpal wall is called a mesiopulpal line angle.

ii. ***Point angle:*** The angle formed at the junction of three adjoining walls in a cavity preparation is referred to as a point angle. For example, angle formed at the junction of mesial, buccal and pulpal walls is called mesio-bucco-pulpal point angle.

iii. ***Axial line angle:*** Any line angle parallel to the long axis of the tooth is called an axial line angle.

iv. ***Pulpal line angle:*** Any line angle horizontal to the long axis of the tooth is called a pulpal line angle.

v. ***Cavosurface angle:*** Angle formed at the junction of the cavity wall and the unprepared tooth surface is referred to as the cavosurface angle. The junction between the cavity wall and the unprepared tooth surface is referred to as the cavosurface margin.

Angles in an Occlusal Cavity (Class I) (Fig. 2.31)

Line Angles
- Mesiofacial
- Mesiolingual
- Distofacial
- Distolingual
- Faciopulpal
- Linguopulpal
- Mesiopulpal
- Distopulpal

Point Angles
- Mesiofaciopulpal
- Distofaciopulpal
- Mesiolinguopulpal
- Distolinguopulpal

Angles in a Proximo-occlusal Cavity in Posterior Teeth (Class II) (Fig. 2.32)

Occlusal Portion

Line angles
- Faciodistal or faciomesial
- Linguodistal or linguomesial
- Faciopulpal
- Linguopulpal
- Mesiopulpal/distopulpal
- Axiopulpal (This line angle is common to both the occlusal and proximal portions of the cavity preparation.)

Point angles
- Distofaciopulpal or mesiofaciopulpal
- Distolinguopulpal or mesiolinguopulpal
- Facioaxiopulpal
- Linguoaxiopulpal

] These point angles are common to both the occlusal and proximal portions of the cavity preparation.

Proximal Portion

Line angles
- Faciogingival
- Linguogingival
- Facioaxial
- Linguoaxial
- Axiogingival

Point angles
- Faciogingivoaxial
- Linguogingivoaxial

Angles in a Proximal Cavity in Anterior Teeth (Class III) (Fig. 2.33)

Line angles

- Faciogingival
- Linguogingival
- Facioaxial
- Linguoaxial
- Axiogingival
- Faciolingual (incisal)

Point angles

- Facioaxiogingival
- Linguoaxiogingival
- Faciolinguoaxial (axioincisal)

Angles in a Proximoincisal Cavity in Anterior Teeth (Class IV)

Incisal Portion

Line angles

- Faciomesial or faciodistal
- Linguomesial or linguodistal
- Faciopulpal
- Linguopulpal
- Mesiopulpal or distopulpal
- Axiopulpal (This line angle is common to both the occlusal and proximal portions of the cavity preparation.)

Point angles

- Faciomesiopulpal or faciodistopulpal
- Linguomesiopulpal or linguodistopulpal
- Facioaxiopulpal
- Linguoaxiopulpal

] These point angles are common to both the occlusal and proximal portions of the cavity.

Proximal Portion

Line angles

- Facioaxial
- Linguoaxial
- Faciogingival
- Linguogingival
- Axiogingival

Point angles

- Faciogingivoaxial
- Linguogingivoaxial

Angles in a Buccal/Lingual Smooth Surface Cavities (Class V) (Fig. 2.34)

Line angles

- Mesioocclusal/mesioincisal
- Distoocclusal/distoincisal
- Mesiogingival
- Distogingival
- Occlusoaxial/incisoaxial
- Gingivoaxial
- Mesioaxial
- Distoaxial

Point angles

- Mesiooccluoaxial/mesioincisoaxial
- Distoocclusoaxial/distoincisoaxial
- Mesiogingivoaxial
- Distogingivoaxial

Angles in Mesio-occluso-distal Cavity (MOD)

Mesial Proximal Box

Line angles

- Facioaxial
- Linguoaxial
- Faciogingival
- Linguogingival
- Axiogingival
- Axiopulpal (This line angle is common to the occlusal and proximal portions of the cavity preparations)

Point angles

- Facioaxiogingival
- Linguoaxiogingival
- Facioaxiopulpal
- Linguoaxiopulpal

] These point angles are common to both the occlusal and proximal portions of the cavity preparation.

Distal Proximal Box

Line angles

- Facioaxial
- Linguoaxial
- Faciogingival
- Linguogingival
- Axiogingival
- Axiopulpal (This line angle is common to both the occlusal and proximal portions of the cavity preparation.)

Point angles

- Facioaxiogingival
- Linguoaxiogingival
- Facioaxiopulpal
- Linguoaxiopulpal

] These point angles are common to both the occlusal and proximal portions of the cavity preparation.

Occlusal Portion

Line angles

- Faciopulpal
- Linguopulpal

BEVEL

Bevel is defined as any angle other than 90°, between the planes or surfaces. Earlier, bevel was placed only on cavosurface margins and defined as "the rounding off of cavosurface margins at an angle".

The bevels being given at various other surfaces of prepared teeth, it is defined as "any abrupt incline between the two surfaces of prepared tooth or between the cavity wall and the cavosurface margins in the prepared cavity".

Bevels are basically given to reduce the marginal errors (space between restoration and tooth surface).

Types of Bevel

a. ***Partial bevel:*** It involves part of the enamel wall (Fig. 2.35A). Such type of preparation is indicated in direct filling gold restorations. A few authors advise giving partial bevel in composite restorations to have more surface area.

b. ***Short bevel:*** It involves the entire enamel wall (Fig. 2.35B). This type of bevel is indicated in cast gold restorations.

c. ***Inverted bevel:*** It is given on the labial shoulder of metal ceramic crowns to effectively improve the esthetics at the margins (Fig. 2.35C).

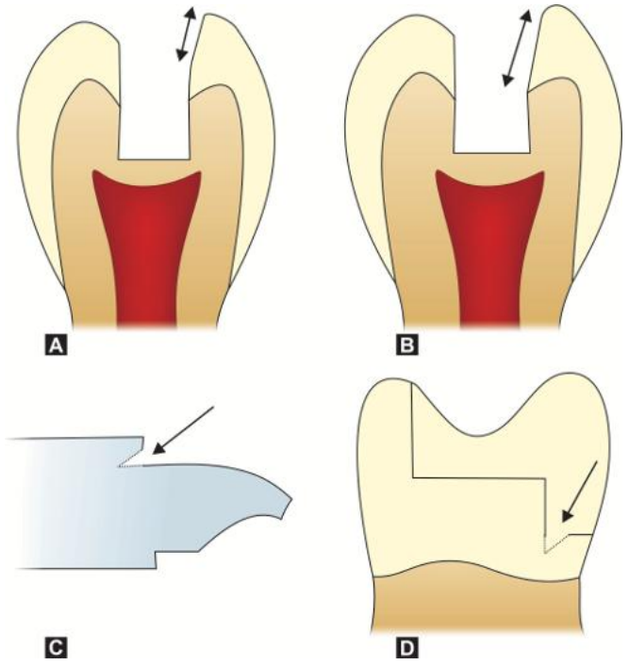


Fig. 2.35: Bevels (arrows): **(A)** Partial bevel; **(B)** Short bevel; **(C)** Inverted bevel; **(D)** Reverse bevel

d. ***Reverse bevel:*** A reverse bevel is placed at the dentinal portion of the cervical wall towards the axiogingival line angle (Fig. 2.35D). The hydrostatic pressure during cementing a cast restoration can produce a rotational displacement of the casting with flat gingival walls. This effect is resisted by the reverse bevel resulting in even seating of the cast restoration.

CLASSIFICATION OF CAVITIES

The first classification of cavities was conceived by Dr GV Black, more than hundred years ago (Table 2.3). It is still being widely used and universally accepted. Though Black originally divided the lesions in five categories; Simon later added the sixth (Fig. 2.36).

Modification of Black's Classification

Black's parameters for classification were controlled by a number of factors; still being followed with slight modifications. Black suggested:

- Removal of tooth structure to gain access and to improve visibility.
- Removal of all traces of affected dentin from the floor of the cavity.
- Provision of mechanical retentive designs.
- Concept of 'extension for prevention'.
- Keeping cavosurface margins at 'self-cleansing' areas.

With the advent of newer adhesive restorative materials, the clinicians started following different

Table 2.3: Classification of cavities (Dr GV Black) Simon later added Class VI cavity

Class I cavities	Cavities beginning in the structural defects like pits and fissures that occur on the occlusal surfaces of premolars and molars, the occlusal two-thirds of the buccal and lingual surfaces of the molars, the lingual surfaces of incisors and any other aberrant locations.
Class II cavities	Cavities in the proximal surfaces of premolars and molars
Class III cavities	Cavities in the proximal surfaces of incisors and canines, but not involving the incisal angle
Class IV cavities	Cavities in the proximal surfaces of incisors and canines but also involving the incisal angle
Class V cavities	Cavities in the gingival third of facial and lingual surfaces of all the teeth
Simon added sixth category as follows:	
Class VI cavities	Cavities on the incisal edges and cusp tips of all the teeth



Fig. 2.36: Class VI (caries involving incisal edges of maxillary teeth)

cavity designs. The concepts given by Black became controversial and debatable. Tunnel preparations and slot preparations were introduced to avoid undue cutting of marginal ridge. However, all these procedures were discarded with time; either because of excessive marginal leakage at the cervical margins or weakening of the marginal ridge. The removal of caries was also not properly accomplished with these procedures.

The knowledge of the presence of two layers of carious dentin, one being infected and the other being affected; and also the phenomenon of remineralization of the affected dentin by fluorides and fluoride releasing restorative materials led to the belief that the affected part of dentin could be left as such.

The mechanical retention design in cavities has been questioned. 'Acid etching followed by bonding provides sufficient retention', was the initial thinking of researchers. The concept was soon challenged and various studies observed that only etching and bonding did not provide sufficient retention. Therefore, though adhesive restorative dentistry has markedly improved the marginal adaptability of restorative materials to cavity walls but these are not ideal substitutes for retention form of cavity as given by Black.

The concept of 'extension for prevention' and keeping the proximal cavosurface margins in 'self-cleansing' areas has also been questioned.

Though Black's classification of cavities/carious lesions is simple, easily followed and universally accepted, certain areas of teeth where caries may occur have been overlooked. Also the spectrum of individual class is a little longer. For example:

i. Carious lesions at line angles of different teeth are not included (Fig. 2.37A, B).
ii. Carious lesions on the labial surfaces of anterior teeth other than in cervical third are not included (Fig. 2.38A, B).
iii. Carious lesions on the lingual surfaces of anterior teeth other than in the cervical third and pits are not included (Fig. 2.39).
iv. Proximal lesions, whether at one side or two sides are taken in one class. MOD cavities are always controversial in Black's classification. A few authors have designated MOD as Class VI without any unanimity for its acceptance.

Mount (1998) has also classified carious lesions/cavities according to site and size (Table 2.4) of the lesion.

The sites are designated as:

- Pit/fissure (1)
- Contact area (2)
- Cervical (3)

Table 2.4: Classification of carious lesions/cavities [Mount (1998)]

Site	*Size*			
	Minimal	*Moderate*	*Enlarged*	*Extensive*
	1	*2*	*3*	*4*
Pit/fissure (1)	1.1	1.2	1.3	1.4
Contact areas (2)	2.1	2.2	2.3	2.4
Cervical (3)	3.1	3.2	3.3	3.4

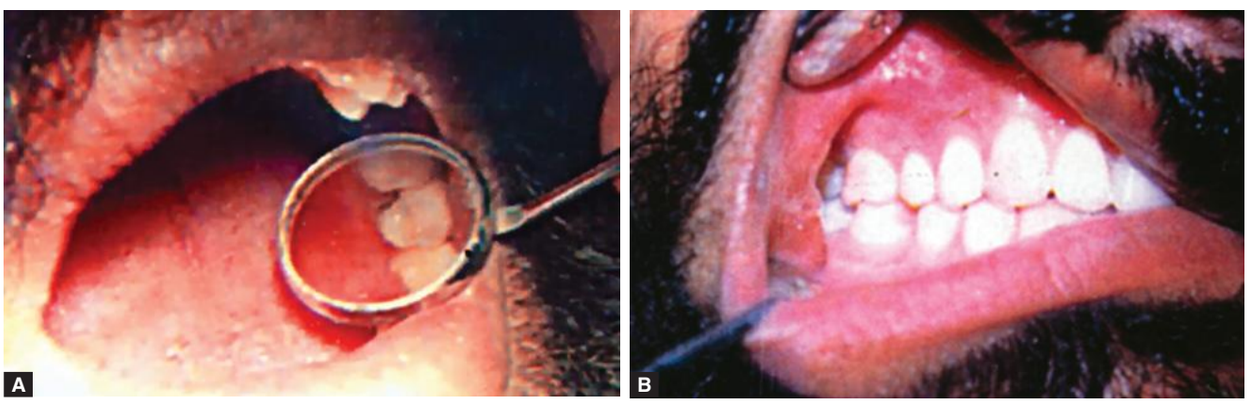


Fig. 2.37: Caries lesions at line angles, not included in Black's classification: **(A)** Caries involving distopalatal line angle of maxillary left first premolar; **(B)** Caries involving mesiobuccal line angle of mandibular right first molar

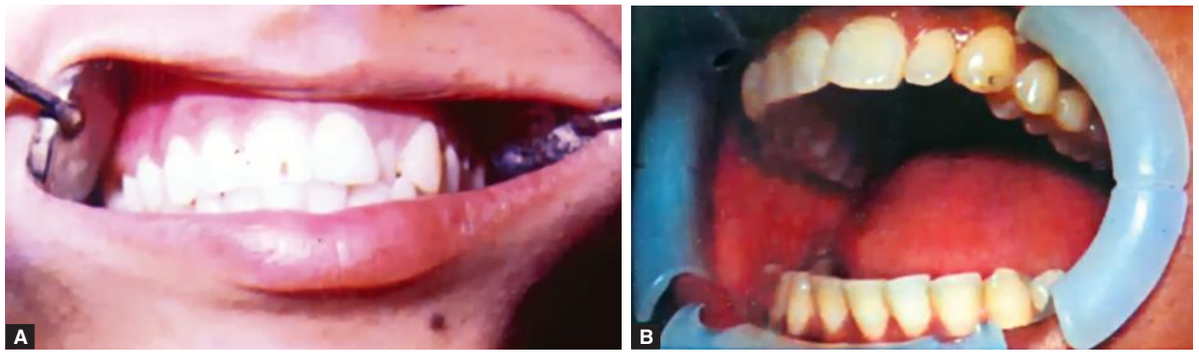


Fig. 2.38: Carious lesions on the labial surface of anterior teeth other than cervical thrid, not included in Block's classification. **(A)** Carious lesion on the labial surface of maxillary right central incisor other than in cervical third; **(B)** Carious lesion on the labial surface of maxillary left canine other than in cervical third



Fig. 2.39: Caries at the cingulum area, not included in Black's classification

The explanation of sizes is:

1. Minimal involvement of dentin. Treatment by remineralization alone.
2. Moderate involvement of dentin. Treatment by cavity preparation. Remaining enamel is sound, well supported by dentin and not likely to fail under normal occlusal load, i.e. the remaining tooth structure is sufficiently strong to support the restoration.
3. The cavity is enlarged beyond moderate size. The remaining tooth structure is weakened to the extent that cusps and incisal edges are split, or are likely to fail, if left exposed to occlusal or incisal load. The cavity needs to be further enlarged so that the restoration be designed to provide support and protection to the remaining tooth structure.
4. Extensive caries with bulk loss of tooth structure has already occurred.

The proposed classification by Mount provides options for treatment planning, keeping in mind treatment by adhesive restorative materials. Though this concept is not entirely unfair, but there is always a subjectivity in deciding the size of the lesion. It may become difficult for an undergraduate to differentiate between different sizes; one-half deciding it as size 3 and the other half as size 4 for the same lesion. Secondly, treatment planning varies with operator-to-operator. Thirdly, the carious sites mentioned earlier which are missing in Black's classification are also missing in this classification. Contact caries (site 2) whether on one side or two sides is taken as one, which is always misleading. And also clubbing root caries with crown caries creates confusion amongst readers.

Keeping in view the simplicity and acceptability of Black's classification, it should not be totally changed; however, little modifications will cover the areas left by Black. A new modified classification, tabulated in Table 2.5, was proposed by the author, Vimal Sikri.

The Class IV cavity of Black is not included in this classification, since with the advent of newer materials and techniques, incisal edge involvement can be managed. The proposed classification does not include root caries. I am of the firm belief that root caries should be dealt separately and in no way be clubbed with the crown caries. Class II Div. 2 clearly shows the involvement of both mesial and distal surfaces of posterior teeth and hence, no confusion with the classification for MOD cavities. The changes are only in the Class IV and Class V cavities proposed by Black. Class V proposed by Black is Class IV Div. 1 in the proposed classification and Class IV Div. 2 are the lesions at the cervical line angles. Class V in this proposed classification is added afresh and includes the lesions, which were not included in Black's classification.

The proposed classification is simple, includes all the surfaces which can be carious and the spectrum of each class is limited. Finally, the record keeping would be easier and the operator's subjectivity has also been taken care of. I believe, the proposed classification would be accepted for future use.

Table 2.5: Classification of cavities [Dr Vimal Sikri (1999)]

Class I	Div. I	Cavities involving pits and fissures of occlusal surfaces
	Div. II	Cavities involving buccal and lingual pits of posterior and anterior teeth
Class II	Div. I	Cavities involving one proximal surface of posterior teeth
	Div. II	Cavities involving both proximal surfaces of posterior teeth
Class III	Div. I	Cavities involving one proximal surface of anterior teeth
	Div. II	Cavities involving both proximal surfaces of anterior teeth
Class IV	Div. I	Cavities on cervical one-third of labial and lingual surfaces of all the teeth
	Div. II	Cavities on labial and lingual line angles of all the teeth
Class V	Div. I	Cavities on labial surfaces of anterior teeth other than cervical one-third
	Div. II	Cavities on lingual surfaces of anterior teeth other than pits and cervical one-third
Class VI	Div. I	Cavities on incisal edges
	Div. II	Cavities on occlusal cusp tips

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